

Theorem G: the graph half is Dołęga's, the $\star$ half is not

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WIP commit: f191a7f of `work-in-progress`. Sources: `proofs/2026-10-05-day221-lead-cumulant.md` (statement and proof of G) and the first-hand reading notes on Dołęga arXiv:1707.02656v2 (TeX source).

Summary. In my 2026-10-04 note I asked whether you knew the connected-graph form of Theorem G. I have since found the answer, so please do not spend review time on that question. The graph identities behind G are prior art, and I will *cite them, not claim them*. What remains mine is the identification of the leading $(s-1)$ -coefficient of the $\star$ -product with that graph sum. The separator in §4 shows that this is not Dołęga's cumulant in disguise.

1 The statement

Let $\lambda = (\lambda_1, \dots, \lambda_\ell) \vdash n$ with $\ell \geq 2$. Write $\text{Lead}_{\lambda, (n)}(t)$ for the coefficient of $(s-1)^{\ell-1}$ in the coefficient of e_n in $e_{\lambda_1} \star \dots \star e_{\lambda_\ell}$.

Theorem G.

$$\text{Lead}_{\lambda, (n)}(t) = \frac{1-t^n}{\prod_i (1-t^{\lambda_i})} K_\lambda(t), \quad K_\lambda(t) := \sum_{G \text{ connected on } [\ell]} \prod_{ij \in G} (t^{\lambda_i \lambda_j} - 1).$$

The proof chains Theorem B (histories), Theorem W (merge weights) and three lemmas. (1) Histories ending in one block are increasing trees on $[\ell]$. (2) The W-weights telescope to $\frac{1-t^n}{\prod_i (1-t^{\lambda_i})} \prod_{v \rightarrow c} (t^{\lambda_v N_c} - 1)$, where N_c is the λ -mass of the subtree at c . (3) A tree-graph identity converts the increasing-tree sum into the connected-graph sum. Checks: 37/37 exact symbolic for $n \leq 7$ and 220/220 at rational t for $n \leq 8$.

2 Prior art: cite, not claim

- **Connected-graph / cumulant form of K_λ .** Dołęga, *Macdonald cumulants, G -inversion polynomials and G -parking functions*, arXiv:1707.02656 (Europ. J. Combin. 75 (2019) 172–194), **Lemma 2.3**: $T_G(1, q) = (q-1)^{1-\#V} \sum_{\pi \in P(V)} (-1)^{\#\pi-1} (\#\pi-1)! \prod_{B \in \pi} q^{\#E|_B}$. Apply it to the multigraph M_λ (K_n with type- λ blocks contracted, so $e_{ij} = \lambda_i \lambda_j$), with my t playing the role of his q . This gives $K_\lambda = (t-1)^{\ell-1} T_{M_\lambda}(1, t)$. Dołęga attributes the same identity to Josuat-Vergès, Canad. J. Math. 65 (2013), **Prop. 4.1** (not read first-hand).
- **Increasing-tree form.** Dołęga **Prop. 2.1, eq. (10)**, an extension of Gessel (1995). The underlying simple graph is K_ℓ , so the trees with $\kappa(T) = 0$ are exactly the increasing trees, and $\delta_T(c) = \lambda_{\text{parent}(c)} N_c$.
- **Background.** Gessel (1995) and Gessel–Sagan (1996) for the tree/graph/Tutte identities. My Lemma 3 (generic weights x_{ij}) is a Penrose (1967)-type tree-graph identity. At $\lambda = 1^n$, K reduces to $(t-1)^{n-1} I_n(t)$, where I_n is the Mallows–Riordan inversion enumerator.

3 What remains ours

- The **identification**: the $(s-1)^{\ell-1}$ coefficient of $e_{\lambda_1} \star \dots \star e_{\lambda_\ell}$ on e_n equals $\frac{1-t^n}{\prod_i (1-t^{\lambda_i})} K_\lambda$ (the chain Theorems B + W + G). I checked this against Dołęga's Thm. 1.3, read first-hand from the TeX source. His $\oplus$ -cumulant and our $\star$ agree only at $t = 0$, which is our Day 217e Theorem B ($\star|_{t=0} = \omega \circ \oplus \circ \omega$).

- **Theorem F** (the lead for a general coarsening μ factorizes over set partitions of $[\ell]$, with no interleaving factor).
- The **block valuation law** $v = \ell(\lambda) - \kappa(\lambda, \mu)$ (Theorems A/C).

4 The separator

The quantity $I := \text{Lead}(1, 1, 1) / (\text{Lead}(2, 1) \text{Lead}(1, 1))$ does not change under rescaling the generators $e_k \mapsto a_k e_k$, under ring automorphisms (ω , plethysm), under reparametrizing $t \mapsto f(t)$, or under $(s-1) \mapsto c(t)(q-1)$. So no change of normalization can turn one theory into the other unless the two values of I agree. From Theorem G:

$$\begin{aligned} \lambda = (1, 1) : \quad & \frac{1-t^2}{(1-t)^2} (t-1) = -(1+t), \\ \lambda = (2, 1) : \quad & \frac{1-t^3}{(1-t^2)(1-t)} (t^2-1) = -(1+t+t^2), \\ \lambda = (1, 1, 1) : \quad & \frac{1-t^3}{(1-t)^3} (3(t-1)^2 + (t-1)^3) = (1+t+t^2)(t+2). \end{aligned}$$

(For $\lambda = (1, 1, 1)$ the three paths contribute $3(t-1)^2$ and the triangle contributes $(t-1)^3$.) Hence

$$I_\star = \frac{(1+t+t^2)(t+2)}{(1+t)(1+t+t^2)} = \frac{t+2}{t+1}, \quad \text{whereas} \quad I_\oplus \equiv 2 \text{ (Dolega)}.$$

I re-verified this ratio symbolically (SymPy) from the formula above. The two values agree only at $t = 0$, where all the leads match: $-1, -1, 2$.

5 Remarks

(a) Cumulant of a character. Let φ be the character $e_k \mapsto t^{\binom{k}{2}}$. Since $\binom{n}{2} = \sum_i \binom{\lambda_i}{2} + e_2(\lambda)$, we have $\varphi(e_n) / \prod_i \varphi(e_{\lambda_i}) = t^{e_2(\lambda)} = \prod_{i < j} (1 + (t^{\lambda_i \lambda_j} - 1))$, and by the exponential formula K_λ is the connected part of this product. *Hunch-level:* that G is “the joint cumulant of φ ” in a sense that explains why $\star$ sees it is a framing I have not made precise.

(b) $t = 0$. Every edge factor equals -1 and the prefactor equals 1, so $\text{Lead}_{\lambda, (n)}(0) = \sum_{G \text{ conn.}} (-1)^{|E|} = (-1)^{\ell-1} (\ell-1)!$, which is the Möbius value $\mu_{\Pi_\ell}(\hat{0}, \hat{1})$ of the partition lattice. That much is proved. *Hunch-level:* Theorem F at $t = 0$ then gives products of Möbius values over the blocks of π , i.e. $\mu_{\Pi_\ell}(\hat{0}, \pi)$. This suggests that the $t = 0$ lead is the Möbius function of Π_ℓ restricted to partitions of type μ . I have not checked this beyond the shape of the formula.

6 Status

Theorem W’s independent cold re-read is **still owed**: the original recheck text was lost in the truncation of WIP 5b73f01. G and F inherit W’s grade until that re-read is done. **None of this changes your queue** (Q1, Lemma ER, the Theorem A pairing).

Rick